

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404089
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Bressan; Michel

Container for ampoules or syringes

Abstract

An improved container for ampoules or syringes has a primary case with at least one opening closable via a primary lid, a tray with a plurality of seats to accommodate the same number of ampoules or syringes, and a secondary lid which defines, inside said primary case, a secondary case which, in turn, accommodates in its interior said tray. When said tray is inserted inside said secondary case, the ampoules or syringes contained therein face towards the inner face of said secondary lid. The secondary lid defines, at each ampoule or syringe below it, an access opening closed by a removable closure portion. Each closure portion has a gripping zone defined by means of a further transverse break line that the user can break by applying pressure at the surface of the gripping zone and more easily remove the closure portion in order to access the underlying ampoule or syringe.

Inventors:	Bressan; Michel (Viggiu', IT)
Applicant:	ALTERGON S.A. (Lugano, CH)
Family ID:	84361545
Assignee:	ALTERGON SA (Lugano, CH)
Appl. No.:	18/704429
Filed (or PCT Filed):	October 25, 2022
PCT No.:	PCT/EP2022/079687
PCT Pub. No.:	WO2023/072885
PCT Pub. Date:	May 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250033865 A1	Jan. 30, 2025

Foreign Application Priority Data

IT	102021000027296	Oct. 25, 2021
KR	10-2021-0186136	Dec. 23, 2021

Publication Classification

Int. Cl.: **B65D85/42** (20060101); **B65D5/50** (20060101); **B65D5/54** (20060101); **B65D75/58** (20060101)

U.S. Cl.:

CPC **B65D85/42** (20130101); **B65D5/503** (20130101); **B65D75/5833** (20130101);

Field of Classification Search

CPC: B65D (5/0227); B65D (5/503); B65D (5/5435); B65D (51/20); B65D (75/5833); B65D (85/42)

USPC: 206/63.3; 206/363; 206/364; 206/369; 206/370; 206/380; 206/438; 206/528; 229/120.21; 229/242; 229/244

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5226532	12/1992	Davidson et al.	N/A	N/A
5762192	12/1997	Jacobs	206/63.5	A61C 7/12
5934460	12/1998	Schmid	206/63.5	A61C 5/66
7424952	12/2007	Antler	229/87.01	A61C 15/00
2005/0121359	12/2004	Lo Duca	206/775	B65D 85/42
2015/0297296	12/2014	Stauder	206/370	B65D 77/0413
2016/0198990	12/2015	Betancur	206/363	B01L 9/06
2024/0150078	12/2023	Marini	N/A	B32B 27/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102010020254	12/2010	DE	N/A
1501985	12/1966	FR	N/A
2734795	12/1995	FR	N/A
2014006207	12/2013	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for Application No. PCT/EP2022/079687, dated Feb. 23, 2023, 10 pages. cited by applicant

Primary Examiner: Bui; Luan K

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is the U.S. national phase of PCT Application No. PCT/EP2022/079687 filed Oct. 25, 2022 which, in turn, claims priority to Italian Patent Application No. 102021000027296 filed Oct. 25, 2021 and Korean Patent Application No. 10-2021-0186136 filed Dec. 23, 2021, the disclosures of which are hereby incorporated in their entirety by reference herein.

FIELD OF THE INVENTION

(2) The present invention relates to an improved container for ampoules, syringes or similar casings containing products, such as for example but not necessarily pharmaceutical products.

(3) The container according to the present invention offers numerous advantages compared to containers of the known type, and in particular is particularly effective in retaining ampoules or syringes in a more stable manner, in allowing identification of individual ampoules or syringes, and in preserving the product contained in said ampoules or syringes from exposure to light, which in the case of many products is harmful.

STATE OF THE ART

(4) Containers configured to hold one or more ampoules or syringes are known on the market, which containers generally comprise a containing case which is usually made of cardboard and opens like a casket and which houses in its interior a tray or box, which may be made of the same material as the containing case or of plastic material, and which in turn comprises a plurality of seats, or blister cells, preformed to hold the same number of ampoules or syringes.

(5) WO 2014/006207 discloses an improved container comprising a primary case provided with an opening, and a tray comprising a plurality of seats for accommodating a plurality of ampoules or syringes according to the pre-characterising portion of claim 1.

(6) However, these containers of the known type have certain disadvantages.

(7) A first disadvantage consists of the fact that with containers of the known type, when the user has to access an ampoule or syringe, they have necessarily to open the containing case which, by opening like a casket, allows access to the ampoules or syringes. In this way, however, all the ampoules or syringes placed inside said tray are exposed to the light.

(8) In many cases, exposure to light of the product, such as a pharmaceutical or cosmetic product for example, is damaging for the product itself.

(9) The same Applicant has carried out some laboratory tests, in particular tests of photostability of pharmaceutical products contained inside ampoules, performed according to the CPMP/ICH/279/95 directive which provides for exposure to the light generated by a fluorescent lamp emitting cold white light according to the specifications of ISO 10977 and by a fluorescent UV lamp with a spectral distribution between 320 and 400 nm with a maximum emission energy comprised between 350 and 370 nm.

(10) By way of example, tests carried out on the proprietary product IBSA Prolutex 25 mg (ratio 2.2) Solution for injection show that the titre results of samples exposed to light show a gradual downward trend compared to the control samples, defined as “dark”.

(11) The same trend was shown for the pH test: the samples exposed to light show a pH value well below that of the dark sample and that of the non-stressed product.

(12) On the basis of the trend found for the progesterone titre test and for the pH test of the solution and considering that both the main pharmacopoeias (EP and USP) lay down that progesterone be stored in containers that protect it from light, it must be concluded that the tested product should be carefully stored in its secondary packaging away from light.

(13) The same obviously applies to many products, particularly but not only pharmaceuticals or cosmetics.

(14) There is therefore a need felt in the sector for having packaging that allows the product contained in ampoules or syringes to be kept away from light until it is used.

(15) A further disadvantage of known containers relates to the fact that, when the user accesses the ampoules or syringes contained therein, opening the containing case, they expose the ampoules or syringes to the risk that they may come out of the housing seats of the tray placed inside the case.

(16) Last but not least, a disadvantage that afflicts containers of the known type consists of the fact that the container does not allow identification of individual ampoules or syringes from among all those contained in the tray.

SUMMARY OF THE INVENTION

(17) In light of the above, the task of the present invention is to overcome the disadvantages that affect containers for ampoules or syringes of the known type.

(18) More particularly, within the scope of this task, an object of the present invention is to provide an improved container for ampoules or syringes capable of eliminating exposure to light of the ampoules or syringes contained therein up to the time when they are used, avoiding exposure to light of all the ampoules or syringes contained therein whenever the lid of the container is opened to take out an ampoule or syringe.

(19) Another object of the present invention is to provide an improved container for ampoules or syringes which allows identification of the individual ampoules or syringes contained therein, in such a way that the user can be guided in the action of withdrawing a specific ampoule, without, however, exposing the other ampoules to light and to the risk of exiting the tray and the container.

(20) A further object of the invention is to provide an improved container for ampoules or syringes, which allows more easily accessing and removing the ampoule or syringe contained therein, without exposing the ampoules to light and to the risk of exiting the tray and the container.

(21) This task and these and other objects are achieved by an improved container for ampoules or syringes according to claim **1**.

(22) Further features of the improved container for ampoules or syringes according to the present invention are the subject of the dependent claims.

Description

LIST OF DRAWINGS

(1) Further features and advantages will be made clearer by the following description by way of a non-limiting example of a preferred embodiment of the present invention which will be given here below with the aid of the accompanying drawings wherein:

(2) FIG. **1** shows the improved container according to the present invention assembled and partially opened;

(3) FIG. **2** shows a perspective view of the tray contained inside the container according to the present invention, which in turn contains, in the example shown, a plurality of syringes;

(4) FIG. **3** shows a plan view of the planar development of the container according to the present invention, without the inner tray.

DETAILED DESCRIPTION OF THE INVENTION

(5) Referring in particular to the accompanying drawings, the improved container **1** according to the present invention comprises a containing case **10** having a substantially box-like shape and provided with at least one opening **11**, preferably a casket-like opening, closable by means of a primary lid **12**, and further comprising a secondary lid **21** which defines, inside said containing case **10**, a box-shaped housing **20** suitable for accommodating a tray or box **30**, advantageously comprising one or more seats, or cells, **31** preformed to accommodate the same number of

ampoules or syringes **50**.

(6) Said tray **30** can be made in the same material as the containing case **10** or in plastic material, according to what is known from the state of the art.

(7) FIG. **3** shows the planar development of the container before it is folded to form the containing case **10** with double lid according to the invention.

(8) Referring in particular to FIG. **1**, said housing **20** defined internally to said primary case **10** is box-shaped and comprises said secondary lid **21** which defines a closure surface that rests above said tray **30** when said tray **30** is inserted in said housing **20**, so that when container **1** is assembled the ampoules or syringes **50** placed inside the cells **31** of said tray **30** face towards the inner face of said secondary lid **21**.

(9) Advantageously, on said secondary lid **21** there are provided, positioned so as to find themselves at the cells **31** which receive said ampoules or syringes **50** when the container is assembled, break lines **22** which each define a closure portion **23** of said secondary lid **21** having a shape substantially corresponding to the underlying cell **31**, so as to define in said secondary lid **21** an access opening **24** sufficiently large to allow the user, once the surface of the secondary lid **21** is broken along said break line **22** and said closure portion **23** is removed, to access the underlying ampoule or syringe **50** so as to be able to take it out.

(10) Advantageously, said containing case **10** and said housing **20** are both made in paper material, preferably in card, even more preferably recycled or environmentally sustainable cardboard.

(11) In this way, it is possible to make the container **1** according to the present invention by folding a planar development shaped according to what can be seen in FIG. **3**, and advantageously made in a single material, for example like said card, in particular recycled card.

(12) Said planar development is first folded to form the housing **20** closed by the secondary lid **21**, and subsequently closed with said primary lid **12** which is superimposed on said secondary lid **21** to form the containing case **10**.

(13) Preferably, said break lines **22** are made up of notches in the secondary lid **21** interrupted by connecting elements that keep the closure portion **23**, defined by the break line **22**, connected to the remaining part of the secondary lid **21**.

(14) Even more preferably, each of said closure portions **23** comprises a gripping zone **25** which is advantageously also defined by means of a further break line **26** arranged transversely with respect to the width of the closure portion **23** so that the user can break the further transverse break line **26** by applying pressure at the surface of the gripping zone **25** and more easily remove the closure portion **23** to take out the ampoule or syringe **50** below.

(15) As a result of what has been described so far, and with reference to FIG. **1**, on the upper surface of the secondary lid **21** of said secondary containing case **20**, which faces towards the exterior when said secondary case **20** is closed, a closure portion **23** is therefore defined at each seat **31** of said ampoules or syringes **50**.

(16) Thanks to this configuration, when the user opens the primary lid **12** of the primary case **10**, they do not have direct access to the ampoules or syringes **50** which are contained, in the seats **31** of said case **30**, inside the secondary containing case **20** which is in turn closed above by said secondary lid **21**.

(17) When the primary lid **12** is opened, in fact, the opening **11** makes the secondary lid **21**, and in particular its upper surface, visible to the user, and there is no direct access to the ampoules or syringes **50**.

(18) In this way, it is ensured that even after opening the lid **12** of the primary case **10**, the ampoules or syringes **50** contained inside the secondary case **20** are not exposed to light.

(19) Furthermore, considering in particular that the lid **12** is reclosable but generally not effectively closed, the configuration of the improved container **1** according to the present invention makes it possible to prevent the ampoules or syringes **50** from accidentally exiting the seats **31** once the lid **12** is opened.

(20) Again, the presence of the secondary lid **21** which is adherent above to the seats **31** keeps the ampoules or syringes **50** stably inside the seats **31** themselves, limiting the movements thereof both when the lid **12** is closed and when it is opened.

(21) Returning to the configuration of the secondary case **20**, as can be seen in particular from the example shown in FIG. 2, on said secondary lid **21** there are present, identified by corresponding break lines **22**, a plurality of closure portions **23** defined by the same number of break lines **22**, each of said closure portions **23** being placed at an ampoule or syringe **50** contained in said tray **30**.

(22) The fact that, after having opened the lid **12** of the primary case, in order to be able to access an ampoule or syringe **50** the user must also remove one of the closure portions **23** formed on the secondary lid **21** so as to have access exclusively to the single ampoule or syringe **50** underlying this closure portion **23**, allows on the one hand, as mentioned, avoiding of exposure to light of all the ampoules or syringes **50** contained in the tray **30** whenever the lid **12** is opened to withdraw an ampoule or syringe **50**, and on the other hand allows affixing on one or more of said closure portions **23** of said secondary lid **21** an identification mark **27** such as a numerical figure, as in the example of FIG. 1, or a letter, or to use a colour code to distinguish and identify specific ampoules or syringes **50** placed below the relative closure portion **23**.

(23) The improved container **1** for ampoules or syringes **50** according to the present invention thus allows identification of the single ampoules or syringes **50** contained therein, so that the user can be guided in the action of withdrawing a specific ampoule without, however, exposing the other ampoules to light and to the risk of them exiting from the tray **30** and from the container itself.

(24) This may be particularly useful when the user has to extract from the container **1** a specific ampoule or syringe, for example to indicate the day of the week so that the user has immediate confirmation that they have taken the product every day, or in the case wherein they have to take different drugs at different times of the day, by resorting for example to a colour code, or similar circumstances.

(25) It has thus been shown how the improved container **1** according to the present invention achieves the intended task and the objects set.

(26) The container **1** according to the present invention may undergo modifications and variations, all of which fall within the scope of the invention as defined by the appended claims.

Claims

1. Improved container for ampoules or syringes comprising a primary case provided with at least one opening closable by means of a primary lid, and a tray comprising a plurality of seats configured to accommodate the same number of ampoules or syringes, further comprising a secondary lid which defines, inside said primary case, a secondary case which, in turn, houses in its interior said tray, so that when said tray is inserted in said secondary case the ampoules or syringes contained therein face towards the inner face of said secondary lid, on said secondary lid being defined, at each ampoule or syringe below it, an access opening closed by a removable closure portion, characterised in that each of said closure portions comprises a gripping zone defined by means of a further transverse break line arranged transversely with respect to the width of the closure portion so that the user can break said further transverse break line by applying pressure at the surface of the gripping zone and more easily remove the closure portion in order to access the underlying ampoule or syringe.
2. Improved container according to the claim 1, characterised in that at least one of said closure portions comprises an identification mark or a colour code designed to distinguish it from the other closure portions.
3. Improved container according to claim 1, characterised in that each of said closure portions comprises an identification mark or a colour code designed to distinguish it from the other closure portions.

4. Improved container according to claim 3, characterised in that said primary case is box-shaped and said opening closable by means of said primary lid is a casket-like opening.
 5. Improved container according to claim 1, characterized in that said secondary case is of a box shape suitable for being inserted inside said primary case.
 6. Improved container according to claim 1, characterised in that on said secondary lid are provided, at the cells that accommodate said ampoules or syringes housed in the tray contained in said secondary case closed above by said secondary lid, break lines that define the shape of said closure portions of said access openings.
 7. Improved container according to claim 6, characterised in that said access openings are of such a size as to allow the user, once said break line has been broken and said closure portion defined thereby has been removed, to access the ampoule or syringe below so as to be able to take it out.
 8. Improved container according to claim 1, characterised in that said break lines are constituted by notches in said secondary lid interrupted by connecting elements that maintain the closure portion connected to the remaining portion of said secondary lid.
 9. Improved container according to claim 1, characterised in that said secondary case and said primary case are made in card.
-